Method of Separation of Variables

In this method, we assume that the solution is the product of two functions, each of which involves only one of the variable.

Questions

1. Solve: $\frac{\partial u}{\partial x} = 3\frac{\partial u}{\partial t}$ using method of separation of variables.

Sol^n:

The given eq^n. is,

$$\frac{\partial u}{\partial x} = 3\frac{\partial u}{\partial t} \quad \text{———} (1)$$

Let, $u = U = XT$ ——— (2) be the solution of eq^n. (1), where, $X = X(x)$, $T = T(t)$

Now, differentiating eq^n. (2) partially w.r.to x,

$$\frac{\partial U}{\partial x} = T\frac{\partial X}{\partial x}$$

Again, differentiating eq^n. (2) partially w.r.to t,

$$\frac{\partial U}{\partial t} = X\frac{\partial T}{\partial t}$$

Now, putting these in eq^n. (1),

$$T\frac{\partial X}{\partial x} = 3\left(X\frac{\partial T}{\partial t}\right)$$

$$\Rightarrow T\frac{\partial X}{\partial x} - 3X\frac{\partial T}{\partial t} = 0$$

Dividing the above eq^n. by XT,

$$\frac{1}{X}\frac{\partial X}{\partial x} - \frac{3}{T}\frac{\partial T}{\partial t} = 0$$

$$\Rightarrow \frac{1}{X}\frac{\partial X}{\partial x} = \frac{3}{T}\frac{\partial T}{\partial t} = k \text{ (say)}$$

Now, $\frac{1}{X}\frac{\partial X}{\partial x} = k$

$$\Rightarrow \frac{1}{X}\partial X = k\,\partial x$$

$$\Rightarrow \int \frac{1}{X}\,dX = \int k\,\partial x$$

$$\Rightarrow \log X = kx + C_1$$

$$\Rightarrow X = e^{kx+C_1} \quad \text{——— (3)}$$

Similarly, $\frac{3}{T}\frac{\partial T}{\partial t} = k$

$$\Rightarrow \frac{3}{T}\partial T = k\,\partial t$$

$$\Rightarrow \int \frac{3}{T}\, \partial T = \int k\, \partial t$$

$$\Rightarrow 3 \log T = kt + c_2$$

$$\Rightarrow T = e^{\frac{kt + c_2}{3}} \quad \text{——— (4)}$$

Now, eqn. (2) becomes,

$$u = e^{kx + c_1} \times e^{\frac{kt + c_2}{3}}$$

$$\Rightarrow u = e^{kx + kt/3} \cdot e^{c_1 + c_2/3}$$

$$\Rightarrow u = B e^{kx + kt/3} \quad [B = e^{c_1 + c_2/3} = e^{c_1 + c_2}]$$

This is the solution of given P.D.E.

2. Use the method of separation of variables to solve the equation $\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u$, given that $u(x,0) = 6e^{-3x}$.

Sol^n:

The given P.D.E. is,

$$\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u \text{ ——— } (1)$$

Let, $u = XT$ —(2) be the solution of above equation.

Now, differentiating eq^n. (2) w.r. to x, we get,

$$\frac{\partial u}{\partial x} = T\frac{\partial X}{\partial x}$$

Again, differentiating eq^n. (2) w.r. to t, we get,

$$\frac{\partial u}{\partial t} = X\frac{\partial T}{\partial t}$$

Now, putting the value of above two in eq^n. (1),

$$T\frac{\partial X}{\partial x} = 2X\frac{\partial T}{\partial t} + XT$$

Dividing above eq^n. by XT,

$$\frac{1}{X}\cdot\frac{\partial X}{\partial x} = \frac{2}{T}\cdot\frac{\partial T}{\partial t} + 1 = K \text{ (Say)}$$

Then, $\frac{1}{X}\frac{\partial X}{\partial x} = K$

$$\Rightarrow \frac{\partial X}{\partial x} = KX$$

$$\Rightarrow \left(\frac{\partial}{\partial x} - K\right) X = 0$$

$$\Rightarrow (D - K)X = 0 \qquad \left[\frac{\partial}{\partial x} \equiv D\right]$$

The A.E. is,

$$D - K = 0$$

$$\Rightarrow D = K$$

$\therefore X = C_1 e^{Kx}$

Also, $\frac{2}{T} \cdot \frac{\partial T}{\partial t} + 1 = K$

$$\Rightarrow 2\frac{\partial T}{\partial t} + T = KT$$

$$\Rightarrow \frac{\partial T}{\partial t} + \frac{T}{2} - \frac{KT}{2} = 0$$

$$\Rightarrow [D' + 1/2 - K/2]T = 0$$

$$\Rightarrow D' - (K/2 - 1/2) = 0 \qquad [A.E.]$$

$$\Rightarrow D' = K/2 - 1/2$$

$\therefore T = C_2 e^{(K/2 - 1/2)t}$

Putting the values of X & T in eqn. (2),

$$u = C_1 e^{kx} \times C_2 e^{(k/2 - 1/2)t}$$

$$\Rightarrow u = C e^{kx} . e^{\left(\frac{k-1}{2}\right)t} \quad \text{--- (3)} \quad [C = C_1 \times C_2]$$

Given, $u(x, 0) = 6 e^{-3x}$

$$\Rightarrow C e^{kx} . e^{\left(\frac{k-1}{2}\right) \times 0} = 6 e^{-3x}$$

$$\Rightarrow C e^{kx} = 6 e^{-3x}$$

Comparing, we get,

$$C = 6 \ \& \ k = -3$$

Putting in eqn. (3),

$$u = 6 e^{-3x} . e^{\left(\frac{-3-1}{2}\right)t}$$

$$\Rightarrow u = 6 e^{-3x} . e^{-2t}$$

$$\therefore u = 6 e^{-(3x + 2t)}$$

This is the required solution of given P.D.E.